GUQINAGENT: Tool-Interactive Performance-Action Planning for Jianpu-to-Jianzipu Generation

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Abstract

Jianzipu, the action-oriented notation of the Chinese guqin, jointly encodes strings, left-hand positions, right-hand strokes, ornaments, and stateful performance conventions. Generating it from a melodic score is therefore not mere text transduction: an output must agree with pitch, tuning, inherited state, and basic performability. We introduce GUQINAGENT, a tool-interactive formulation of constrained jianzipu generation. Our contributions are: (i) a full-score benchmark and executable runtime for iteratively editing a jianpu score into jianzipu; (ii) a warning-guided inference mechanism that feeds deterministic pitch and conservative playability diagnostics back to the agent without forcing a rewrite; (iii) a two-stage agent that separates pitch-grounded fingering from idiomatic guqin refinement; and (iv) targeted inverse trajectory construction that converts annotated scores into auditable, redacted tool-use supervision. Phrase partitioning is part of the generation algorithm, not an assumption on the input. The runtime exposes pitch candidates, stateful editing, compact context retrieval, and advisory warnings, and supports structured event-level evaluation rather than string overlap. We will release the model, redacted training trajectories, test benchmark, and executable runtime. On an initial evaluation of 48 scores, the full system achieves 99.981% pitch accuracy within 50 cents on 15,758 auditor-resolvable events. Structured action and ornament scores remain substantially lower, motivating evaluation of idiomatic realization alongside pitch correctness.

1 Introduction

Jianzipu is an open, compositional notation system that encodes how a guqin performance is produced, rather than merely which pitches sound. A notation unit can combine a string, hui position, left-hand finger, right-hand stroke, and ornament in one compact action description. Its interpretation is also stateful: harmonic mode remains active until terminated, walking gestures inherit positions, and compound gestures may cover multiple events. Similar text can thus imply different pitches, while different surface forms can be musically equivalent (Li et al., 2026b,a).

Recent work has made important progress on the preceding digitization problem: recognizing jianzi from manuscript images and recovering their internal compositional structure, including zero-shot component-sequence recognition and full-page mixed-script extraction (Li et al., 2026b,a). Our problem begins after such recognition. Given a machine-readable jianpu melody and score metadata, the system must select an executable realization in jianzipu. This is not transcription: the input does not prescribe a unique string, hui, finger, or ornament. It is a constrained performance-action planning problem, where pitch agreement is necessary but not sufficient.

Existing text generation alone does not expose whether a candidate realizes the intended pitch under the current tuning, inherits harmonic or left-hand state correctly, or remains compatible with notation coverage rules. Recent language-model research has explored interleaving reasoning with external actions, API use, and iterative feedback, but those methods do not model guqin-specific tuning, notation state, or performance constraints (Yao et al., 2023; Schick et al., 2023; Madaan et al., 2023). We formulate generation as interactive full-score plan editing. The model observes a complete jianpu score, partitions it into context-aware work units, queries deterministic tools when useful, and incrementally writes one event-indexed final notation plan. Pitch and state inconsistencies are exposed as actionable feedback, while idiomatic choices remain model decisions. In summary, our contributions are fourfold:

1. We introduce *constrained jianzipu generation*,

a benchmark and executable runtime for reproducible score-to-jianzipu evaluation.

2. We introduce warning-guided inference: deterministic pitch and conservative notation-constraint diagnostics become visible, actionable feedback rather than hidden post-hoc validation.

3. We design GUQINAGENT, a two-stage Fingering→Guqinizer agent that distinguishes basic sound production from stylistic refinement.

4. We develop targeted inverse trajectory construction, which replays annotations through the runtime and exports public, redacted supervision.

2 Related Work

Guqin notation and performance-oriented symbolic music. Recent work on jianzipu has focused on recovering structured notation from glyphs and manuscript pages (Li et al., 2026b,a). Earlier language-model-based work also studied transcription from jianzipu into modern notation (Shi et al., 2024). Our setting reverses this direction: starting from melodic events, the system must choose how to realize them on the instrument. A related problem arises in guitar tablature inference, where a pitch may admit several string–fret assignments and playability is assessed by musicians (Edwards et al., 2024). Work on piano fingering likewise models hand position and execution difficulty (Iman and Ahn, 2024). At the composition level, YuE2 generates a readable melody-and-chord score before synthesizing full-song audio; the score supports direct editing and revisions driven by external language models in response to user feedback (Yuan et al., 2026). GUQINAGENTstudies how a given score is realized through guqin performance actions, including state-dependent gestures and compact notation whose interpretation depends on tuning and prior actions.

Tool-interactive planning and revision. ReAct interleaves reasoning and environment actions, while Toolformer and ToolLLM investigate tool-use learning and data construction (Yao et al., 2023; Schick et al., 2023; Qin et al., 2024). Self-Refine and Reflexion use iterative feedback to improve model behavior; CRITIC specifically studies revision driven by external tool outputs (Madaan et al., 2023; Shinn et al., 2023; Gou et al., 2024). GUQINAGENTuses a shared executable notation state to check pitch realizations and issue conservative, event-level warnings. These checks inform subsequent edits but do not determine musical suitability on their own.

Learning from interaction trajectories. AgentTuning and FireAct demonstrate the value of supervised interaction trajectories for language agents (Zeng et al., 2024; Chen et al., 2023). More closely related, RetroGen reconstructs process supervision from final expert artifacts for evidence-grounded generation (Huang et al., 2026). Our teacher likewise has access to a target artifact during offline construction, but is required to execute edits in the student-facing guqin runtime; the retained training context contains only public actions and observations. GEPA instead optimizes prompts by reflecting on execution feedback without updating model weights, providing a distinct comparison to trajectory fine-tuning (Agrawal et al., 2026).

3 Methodology

3.1 Constrained Generation with an Executable Runtime

Given a complete jianpu score $x = (m, e_{1:N})$, the task is to construct an event-indexed jianzipu plan $y = (y_1, \ldots, y_N)$. The metadata m specifies the mode and tuning; each event e_i has a stable index, notated duration, structural type, and a target pitch q_i when defined. An output entry y_i is either a jianzipu expression or an explicit empty display value. The latter may indicate continuation or coverage by a compound gesture and does not remove the event from the score. ABC notation is used only as an internal representation for pitch processing, not as an additional input.

Unlike independent token prediction, the interpretation of y_i depends on preceding actions. We represent this dependence by a deterministic notation-state transition,

$$s_i = \mathcal{T}(s_{i-1}, y_i; m), \qquad s_0 = s_{\text{init}}(m), \quad (1)$$

where s_i records such properties as harmonic-region status, inherited left-hand position, walking gestures, and compound-action coverage. Given this state, the realized pitch $\hat{q}_i = \mathcal{P}(y_i, s_{i-1}; m)$ is a partial function: some compound or ambiguous expressions do not admit a reliable pitch estimate. Structural consistency and pitch agreement

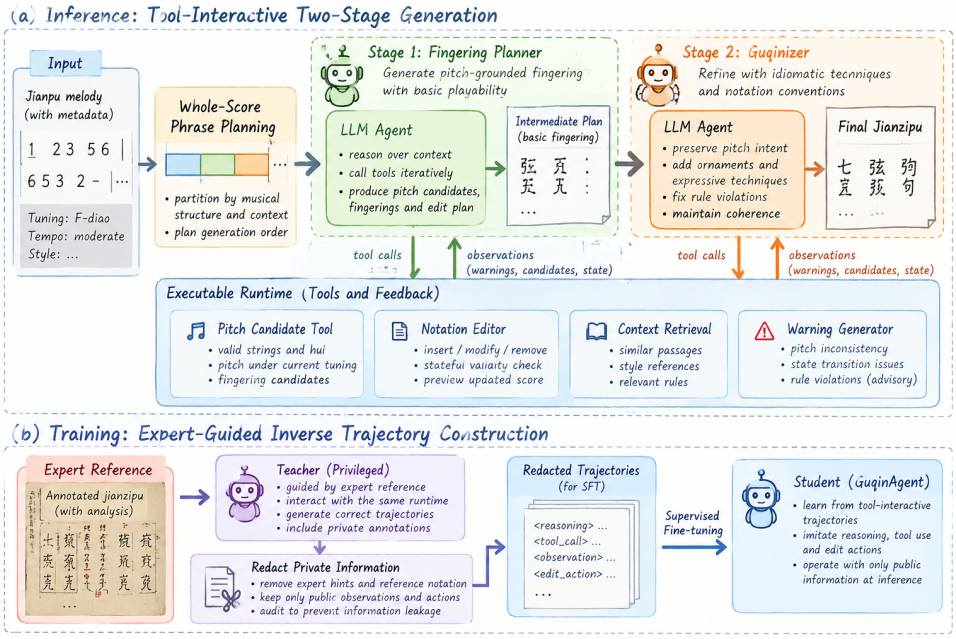


Figure 1: Overview of GUQINAGENT. **Top:** at inference time, the agent partitions a complete jianpu score into ordered work units, constructs a pitch-grounded intermediate plan in the Fingering stage, and refines it into final jianzipu in the Guqinizer stage. Both stages interact with the same executable runtime, which returns pitch candidates, plan previews, compact context, and advisory warnings. **Bottom:** for training, a privileged teacher uses expert annotations only through a private channel while interacting with the public runtime. Redaction retains a student-visible trajectory for supervised fine-tuning, so the deployed student conditions only on score context, its prior actions, and runtime observations.

are therefore distinct from the broader question of musical suitability.

Interactive plan editing. The complete score is partitioned into an ordered sequence of bounded, musically aligned work units $\mathcal{B}(x) = (p_1, \ldots, p_K)$. This partition is computed from the input score; the benchmark examples remain complete scores. The agent edits one shared plan rather than generating independent phrases. At interaction step t within p_k, its public history consists of the score, current work unit, editable plan, replayed notation state, and previous tool observations:

$$h_t = (x, p_k, y^{(t)}, s^{(t)}, o_{<t}),$$
$$a_t \sim \pi_\theta(\cdot \mid h_t, r), \tag{2}$$

where r denotes the current planning stage and $s^{(t)}$ is obtained by replaying the current plan. The executable runtime $\mathcal{R}$ processes an action and returns the resulting plan, state, and tool observation:

$$(y^{(t+1)}, s^{(t+1)}, o_t) = \mathcal{R}(x, y^{(t)}, s^{(t)}, a_t). \tag{3}$$

Read-only actions leave the plan unchanged. The runtime used to construct training trajectories is also used at inference time.

The agent can request tuning-aware string–hui realizations through `get_pitch_candidates`, including open strings, stopped notes, harmonics, and supported inherited or compound gestures. Returned candidates distinguish exact, approximate, and unavailable matches. The only plan-writing operation is `edit_plan(jianzi_rows)`, which applies event-indexed updates and replays the resulting notation state. Compact context retrieval through `list_context` and targeted earlier-phrase lookup provides prior material without reproducing the entire score in every interaction.

Conservative constraint feedback. We distinguish runtime-checkable violations from cases whose interpretation is uncertain. For a rule c and event i, let $v_c(i; x, y, s) \in \{0, 1, \perp\}$ denote no detected violation, a detected violation, or an indeterminate check, respectively. The warning set is

$$W(x, y) = \{(i, c) : v_c(i; x, y, s) = 1\}, \tag{4}$$

with s obtained by replaying y. In the implementation, checks cover reliably parsed pitch mismatches and conservative diagnostics for harmonic-state

continuity, walk direction or span, interrupted open-string resonance, paired stroke adjacency, and difficult left-hand reaches. Indeterminate cases are not converted into violations. Warnings accompany the relevant `edit_plan` preview but neither rewrite the plan nor imply that an unflagged output is musically correct. The agent may query candidates, revise the affected entries, or retain its choice; identical repeated edits are stopped to prevent unproductive loops.

3.2 Two-Stage Performance-Action Planning

The same notation event jointly specifies how a pitch is produced and how it is articulated. We separate these decisions into two successive planning stages. *Fingering* first constructs a basic pitch-grounded realization, choosing strings, hui positions, left-hand contact, and elementary right-hand strokes. *Guqinizer* then revises the committed plan for idiomatic execution, including ornaments, walking gestures, harmonic transitions, and compound-action notation. The second stage may preserve or revise earlier decisions; it is not a post-processing formatter.

For work unit p_k, let $y^{(k-1)}$ be the shared plan after completing the preceding units. A bounded runtime rollout under each stage gives

$$\begin{aligned} y^{F,(k)} &= \mathrm{Rollout}_F(x, p_k, y^{(k-1)}; \mathcal{R}), \\ y^{(k)} &= \mathrm{Rollout}_G(x, p_k, y^{F,(k)}; \mathcal{R}), \end{aligned} \tag{5}$$

where each rollout consists of stage-conditioned tool interactions of the form in Eq. (3). Only edits committed through the runtime enter the shared plan. Crucially, the state passed to p_{k+1} is replayed from $y^{(k)}$, including any earlier mistakes, rather than substituted from an expert reference. Both stages retain access to pitch candidates and runtime feedback. Warnings can trigger another revision but are not hard acceptance criteria; execution remains subject to the configured interaction budget.

This decomposition imposes a procedural bias: establish a basic sound-producing realization before making the more context-dependent choices involved in idiomatic execution. It does not assume a unique separation between fingering and ornamentation, nor does it guarantee that the second stage preserves every correct decision. We evaluate both the benefits and regressions introduced by refinement against a single-stage agent using the same public tools.

3.3 Expert-Guided Inverse Trajectory Learning

Expert jianzipu annotations specify desired performance actions but generally not the tool interactions by which an agent could arrive at them. Directly supervising final notation therefore leaves the interaction policy unspecified. We instead construct trajectories by allowing an offline teacher to work backward from the annotated target while acting through the same executable runtime available to the student.

Let z be a phrase-level expert annotation and $\mathcal{K}(z)$ the technique-relevant descriptions retrieved from a curated knowledge base. Both are private to the teacher. Its rollout is conditioned on the input, private guidance, and public runtime:

$$\tau^T \sim \mathrm{Rollout}\big(\pi_T \mid x, z, \mathcal{K}(z); \mathcal{R}\big), \tag{6}$$

where τ^T records the chronological sequence of tool actions, returned observations, and accepted plan updates. The reference guides the search for an executable realization but does not constitute a tool response. The teacher constructs stage-specific traces: basic fingering first, followed by idiomatic refinement. Semantic descriptions clarify technique constraints that may be implicit in compact jianzipu symbols; they are not added to the student's inference context.

Validation and public-prefix reconstruction. Candidate examples are screened for reference parsing and consistency before construction, and generated traces are checked against the runtime. Training examples are obtained by removing private expert content and retaining the public interaction sequence,

$$\begin{aligned} \tilde{\tau} &= g_{\mathrm{pub}}(\tau^T), \\ \mathcal{D} &= \{\tilde{\tau} : \tau^T \text{ passes validation}\}. \end{aligned} \tag{7}$$

The intended information boundary is temporal as well as structural: at each supervised action, the conditioning context should contain only the input, earlier public actions, and observations already returned by the runtime. Removing private fields alone cannot establish this property if rewritten reasoning anticipates future observations or reproduces unseen annotation details. Public-prefix consistency must therefore be checked separately from runtime execution validity; the precise checks and filtering counts belong in the experimental protocol.

Component	Contents
Input	Complete-score metadata, tuning, jianpu events, and stable event indices.
Output	One event-indexed jianzipu plan for the complete score.
Runtime	Pitch candidates, editing, context retrieval, validation, and diagnostic feedback.
Reference	Expert notation for offline supervision and structured action comparison.

Table 1: Benchmark representation. Reference notation is withheld from inference.

Student supervision. We fit a stage-conditioned student policy to the assistant-side actions and messages in the retained public trajectories. For a supervised token position u, let h_u^{pub} contain the preceding student-visible context and let w_u be its target token. The training loss is standard masked cross-entropy,

$$\mathcal{L}_{\text{SFT}}(\theta) = -\sum_{\tilde{\tau}\in\mathcal{D}} \sum_{u\in\mathcal{Y}(\tilde{\tau})} \log p_\theta(w_u \mid h_u^{\text{pub}}), \quad (8)$$

where $\mathcal{Y}(\tilde{\tau})$ indexes supervised assistant tokens; tool outputs form context but are not prediction targets. At deployment, neither expert annotations nor private semantic guidance are provided. The resulting student must recover its actions from the input score, its own previous edits, and the executable feedback it has observed.

4 Experiments

4.1 Experimental Setup

Benchmark and Data Protocol. The current evaluation contains 48 complete scores, partitioned into 1,031 work units, with 23,602 aligned event records. It includes a replacement edition of *Youlan* containing 52 work units; three problematic source scores have been excluded. Scores are the units of dataset splitting, while phrases are internal work units used by the agent. Both currently evaluated conditions produce protocol-valid outputs for all 1,031 work units.

All outputs are evaluated using the same current rules after replaying each complete score in chronological order. This preserves harmonic regions, left-hand positions, and other state across work-unit boundaries. Runtime snapshots and evaluation code fingerprints are retained for auditability. The final reproducibility record will specify the split manifest, checkpoint, training configuration, and inference arguments. [TODO: Add training-set size, adapter checkpoint hash, SFT hyperparameters, and inference budgets.]

Evaluation Metrics. Pitch correctness is measured against the input jianpu melody. The expert jianzipu annotation supplies reference actions for fingering and technique metrics. Let $\mathcal{I}$ contain events for which the auditor returns a determinate matched or mismatched prediction. Let $a_i = 1$ if event i matches the target within 50 cents under the supported matching rules, and $a_i = 0$ otherwise. Let $\mathcal{A}_i$ contain the tone pairs aligned by the auditor, with absolute error δ_{ij} in cents for pair j. We report

$$\begin{aligned} \text{PitchAcc}_{50} &= \frac{\sum_{i\in\mathcal{I}} a_i}{|\mathcal{I}|}, \\ \text{PitchMAE} &= \frac{\sum_{i\in\mathcal{I}} \sum_{j\in\mathcal{A}_i} \delta_{ij}}{\sum_{i\in\mathcal{I}} |\mathcal{A}_i|}. \end{aligned} \quad (9)$$

Accuracy uses events as its denominator; MAE averages aligned tone pairs, and a compound event can contribute multiple pairs. Unresolved cases are excluded from pitch scoring and must be reported separately. We report each condition's evaluable event count and accuracy on their intersection, since generated notation affects which expressions the auditor resolves. The complete event inventory includes structural and continuation events and is not the pitch denominator.

String, left-hand fingering, right-hand fingering, and tone type are compared as structured fields against available reference labels using micro precision, recall, and F1. Tone type distinguishes open, stopped, and harmonic notes. Ornament precision, recall, and F1 use event-level sets of parsed ornaments. These metrics measure agreement with one expert realization; alternative playable realizations can receive lower scores. Technique distributions provide a complementary view of underused and overused actions.

Final warnings are counted by replaying completed plans through the runtime, including when warnings were hidden during inference. The rule-violation rate is the fraction of the 1,031 work units with at least one final warning; the total warning count is also reported. Unresolvable expressions do not automatically become violations, and a warning-free plan is not certified as playable.

Models and Comparisons. The current results use Qwen3.5-9B with the same project-trained two-stage LoRA weights in both conditions. The full system exposes warnings, diagnostic hints,

Method	Valid coverage	Pitch Acc. ↑	MAE ↓	String F1	LH F1	RH F1	Tone F1	Orn. F1
Original Qwen3.5-9B, direct	–	–	–	–	–	–	–	–
Original Qwen3.5-9B, agent†	1.07	N/A	N/A	N/A	N/A	N/A	N/A	N/A
GLM API, agent + GEPA	–	–	–	–	–	–	–	–
Project SFT, single-stage agent	–	–	–	–	–	–	–	–
Project SFT, two-stage, feedback OFF	100.00	86.237	64.993	38.554	56.599	45.659	61.670	22.285
Project SFT, two-stage, feedback ON	100.00	99.981	2.308	39.761	57.119	46.009	62.543	20.184

Table 2: Preliminary results on 48 scores. Coverage, accuracy, and F1 are percentages; MAE is in cents. Valid coverage is protocol-valid outputs divided by all 1,031 test work units. Pitch accuracy uses each condition's evaluable events (Table 3); action F1 compares with expert notation. †The original-model agent produced valid outputs for 11/59 attempted units; 972 units were skipped after predecessor failures. N/A denotes unavailable full-test musical metrics; – denotes pending experiments. ON and OFF use the same adapted backbone and scoring rules.

and warning-triggered follow-up interactions. The diagnostic-feedback-off condition hides this feedback and disables its follow-up mechanism while retaining structural validation and pitch candidates. Its public user and tool messages were checked for diagnostic-feedback leakage.

We also evaluated the original Qwen3.5-9B checkpoint without the project LoRA using the two-stage agent and the same tool definitions. All 59 attempted inputs matched the test-input hashes. The original-model run allowed 8,192 output tokens per request, eight rounds per stage attempt, up to three attempts, and up to three Guqinizer passes, with a context limit of 16,384 tokens. After a work unit failed, the evaluator skipped the remainder of that score, preserving the requirement that history contain model-generated predecessors. These settings match the current SFT evaluation entry point; archived SFT outputs span multiple runs, so this comparison does not establish that every historical execution setting was identical.

The remaining comparisons examine interactive generation, stage decomposition, and a commercial-model baseline. Direct generation with the original checkpoint is an optional additional comparison. The GEPA baseline will use a commercial GLM model through its API as the agent backend, with public prompts optimized on development data without updating model weights (Agrawal et al., 2026). Differences from the Qwen student reflect both the model backend and the training or prompt-optimization approach. The GLM model version, GEPA reflection model, API settings, optimization budget, and inference costs will be reported. A student trained on single-stage trajectories will generate final jianzipu through the same tools; training and inference token budgets will be reported.

4.2 Results and Discussion

Original-model completion and protocol failures. The original-model agent attempted 59 of 1,031 work units across the 48 scores. Eleven attempts produced protocol-valid outputs (18.64% of attempted units), while 48 failed. The remaining 972 units were skipped because a predecessor failed. Valid-output coverage was therefore 11/1,031 (1.07%); skipped units are excluded from the attempted-unit protocol-failure denominator. Both adapted conditions produced valid outputs for all 1,031 units. We do not report full-test musical metrics for the original model or compare its 11 successful units directly with the adapted model's complete-test scores.

Of the 48 failed units, 13 failed during Fingering and 35 during Guqinizer. Logs record output truncation in 45 failed units, at least one attempt reaching round eight in 26, and three consecutive pitch-candidate-query rounds in 31. These observations overlap and are not a mutually exclusive attribution of failures. They suggest that sustained tool interaction and termination behavior are substantial obstacles for the original model under the current protocol, alongside the musical decisions required by the task.

Pitch correctness and diagnostic feedback. The full system attains 99.981% pitch accuracy on 15,758 evaluable events, compared with 86.237% on 15,360 events when diagnostic feedback is disabled (Table 2). MAE is 2.308 versus 64.993 cents. Since the evaluable sets differ, Table 3 additionally reports accuracy on 14,289 events resolved in both conditions: 99.979% versus 86.941%, a gap of 13.038 percentage points. The observed accuracy

Diagnostic	ON	OFF
Evaluable pitch events	15,758	15,360
Aligned tone pairs for MAE	16,335	15,958
Common evaluable events	14,289	14,289
Pitch Acc. on common events (%)	99.979	86.941
Final runtime warnings	2	3,436
Work units with warnings	2	809
Rule-violation rate (%)	0.194	78.468

Table 3: Diagnostic-feedback comparison. Counts refer to final plans, rescored under the same runtime with complete histories. Each condition covers 1,031 work units. OFF warnings are measured during evaluation despite being hidden during inference.

gap therefore persists on the same input events. The result concerns auditor-supported realizations and does not establish accuracy on unresolved notation.

Residual warnings and error analysis. The full system leaves two final runtime warnings in two work units, compared with 3,436 warnings across 809 work units for OFF. The affected-work-unit rate is 0.194% versus 78.468%. One remaining ON warning in *Youlan* is an octave error, realizing E6 for a target E7. These cases warrant inspection of inherited state, compound notation, and ordinary pitch choices. Final-warning counts and pitch-error counts use different eligibility and matching rules and should not be treated as interchangeable metrics.

Fingering agreement and idiomatic realization. The full system's string, left-hand, right-hand, and tone-type F1 scores are 39.761, 57.119, 46.009, and 62.543, respectively. Their modest differences from OFF accompany a much larger pitch-accuracy gap. The full system also has lower ornament F1: 20.184 versus 22.285. The current comparison provides evidence of improved pitch consistency without establishing an advantage in ornament selection.

Tone distributions indicate that pitch accuracy leaves room for different realization strategies. Among recognized tone types on reference-labeled events, harmonics account for 7.51% of ON predictions, compared with 29.34% of reference tone types. A system can realize the correct pitch through different strings, positions, or tone types, and can also underuse stylistically appropriate techniques. Both possibilities motivate examining action distributions and consulting performers when assessing idiomatic quality.

Planned expert assessment. A blinded performer study will complement automatic metrics. Proposed dimensions are execution feasibility, continuity of left- and right-hand choices, and idiomatic guqin expression. Examples will be anonymized and presented in randomized order, with input scores available to raters. The study will report rater experience, sampling, agreement between raters, and uncertainty of the estimates. Expert scores and claims of superior playability await this assessment.

5 Conclusion

GUQINAGENTframes jianzipu generation as executable, stateful plan editing. Its benchmark makes score events, tuning, and notation state explicit; its two-stage architecture separates basic realization from idiomatic refinement; and its inverse trajectories create auditable public tool-use supervision from private annotations. Together, these components enable evaluation beyond text overlap and establish a practical basis for musically grounded notation agents. Initial results on 48 scores show high pitch consistency on auditor-resolvable events, while lower fingering and ornament agreement identifies idiomatic realization as a continuing challenge. Comparisons with the original checkpoint, GEPA, and a single-stage student, together with performer assessment, will establish how training, architecture, and diagnostic feedback contribute to the final notation.

6 Limitations, Ethics, and Resource Availability

The auditor is conservative and incomplete: no warning is not proof of musical correctness, and unresolved notation is not a success. Fine-grained left-hand reachability, timbre, and lineage-specific style still require expert review or additional physical modeling. The digitized score collection may not represent all historical sources or playing traditions. We position GUQINAGENTas an assistive research system, not a replacement for performers or tradition bearers. Before release, we will review source licenses and provenance, remove private teacher material, and publish only redacted trajectories intended for public supervision. Since inference feedback and final warning scoring share domain rules, warning reductions are not independent evidence of performer-rated quality. Model-dependent parseability changes the pitch denomi-

nator even with a fixed input score. The pending original-checkpoint, GEPA, single-stage, and expert comparisons are necessary to assess generality and execution quality.

Open Science and Resource Availability. We will release the trained model, redacted training trajectories, the test benchmark, and the executable runtime used by teacher generation and inference. The release will include score-level split definitions, event-indexed schemas, evaluation scripts, filtering manifests, and instructions for reproducing the benchmark. Private teacher prompts, unredacted teacher reasoning, and any source material that cannot be redistributed will not be included. The archival repository URL, model license, dataset license, and versioned release identifier will be added before submission: [TODO: repository URL and licenses].

References

Lakshya A. Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J. Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. 2026. GEPA: Reflective prompt evolution can outperform reinforcement learning. In *International Conference on Learning Representations*.

Baian Chen, Chang Shu, Ehsan Shareghi, Nigel Collier, Karthik Narasimhan, and Shunyu Yao. 2023. FireAct: Toward language agent fine-tuning. *Preprint*, arXiv:2310.05915.

Andrew C. Edwards, Xavier Riley, Pedro Sarmento, and Simon Dixon. 2024. MIDI-to-Tab: Guitar tablature inference via masked language modeling. In *Proceedings of the 25th International Society for Music Information Retrieval Conference*, pages 288–294.

Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. 2024. CRITIC: Large language models can self-correct with tool-interactive critiquing. In *International Conference on Learning Representations*.

Junjie Huang, Jiarui Qin, Di Yin, Weiwen Liu, Yong Yu, Xing Sun, and Weinan Zhang. 2026. From final artifacts to trajectories: Retrospective process supervision for evidence-grounded long-form generation. *Preprint*, arXiv:2608.30461.

Ananda Phan Iman and Chang Wook Ahn. 2024. A model-free deep reinforcement learning approach to piano fingering generation. In *2024 IEEE Conference on Artificial Intelligence (CAI)*, pages 31–37. IEEE.

Zehan Li, Fu Zhang, Zhijun Liu, and Jingwei Cheng. 2026a. End-to-end structured information extraction from mixed-script documents in open compositional symbol systems. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.

Zehan Li, Fu Zhang, Zhijun Liu, and Jingwei Cheng. 2026b. Zero-shot jianzi recognition as structured visual information extraction in open compositional symbolic systems. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 29414–29429.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, volume 36.

Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Lauren Hong, Runchu Tian, Ruobing Xie, Jie Zhou, Mark Gerstein, Dahai Li, Zhiyuan Liu, and Maosong Sun. 2024. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. In *International Conference on Learning Representations*.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*, volume 36.

Kaiwen Shi, Kan Liu, and Xizi Zhang. 2024. Guqin jianzi notation transcription based on language model. In *Music Intelligence: Second Summit, SOMI 2023, Revised Selected Papers*, pages 66–79. Springer Nature Singapore.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R. Narasimhan, and Yuan Cao. 2023. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*.

Ruibin Yuan, Jiahao Pan, Junyan Jiang, Zhiyue Wu, Ziya Zhou, Jiankai Sun, Yizhi Li, Ge Zhang, Yicheng Gu, Zeyue Tian, Junyu Dai, Hanfeng Lin, Kai Li, Shangda Wu, Xuanjie Liu, Jiaming Wang, Zihan Liu, Yue Wang, Yinghao Ma, and 16 others. 2026. YuE2: Unifying symbolic and audio music generation at frontier quality. *arXiv preprint arXiv:2609.33757*.

Aohan Zeng, Mingdao Liu, Rui Lu, Bowen Wang, Xiao Liu, Yuxiao Dong, and Jie Tang. 2024. AgentTuning: Enabling generalized agent abilities for LLMs. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3053–3077. Association for Computational Linguistics.